



**RAJIV GANDHI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 3

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Teacher's Sign :

Department Vision And Mission

Vision

Cultivate student as technocrats researchers and entrepreneurs in the field of computer science and engineering.

Mission

- * The impact quality education in computer science and engineering by means of learning techniques and Value-added courses.
- * To inculcate professional excellence and research culture by encouraging projects in cutting edge technologies, through industry interaction.
- * To build leadership, skills ethical values and team work among the students.
- * To strengthen the collaboration of departments and industry internships, mentorship and professional body activities

PART - A

1. Define DTD?

- A DTD is stands for document type definition.
- DTD is a set of rules that defines the structure and organization of an XML document, including the allowed elements, attributes, and relationships between them.

2. What is the use XML element?

1. Define structure: Organize and structure data in a document.

2. Represent data: Hold and represent data in a format that is easy to read and write.

XML elements are used in:

- data exchange and storage
- RSS feeds
- configuration files
- web services.

3. Give the syntax of DTD?

<!ELEMENT element-name
(content-model)>

<!ATTLIST element-name
attribute-name attribute-type
default-value !>

4. Define XML schema?

XML schema is a language used to define the structure and constraints of an XML document, including:

- element names and relationships
- data types and formats.
- attribute names and data types
- cardinality and sequence of elements.

5) List the data type supported in XML schema?

1. String: character strings
2. Integer: whole numbers.
3. decimal: decimal numbers
4. Boolean: true or false values
5. Date time: dates and times
6. Byte: binary data.

PART-B

1. Explain in details about document type definition?

- A DTD is a document type definition.
- A DTD defines the structures and the legal elements and attributes of an XML document.

Why use a DTD?

With a DTD, independent groups of people can agree on a standard DTD for interchanging data.

An application can use a DTD to verify the XML data is valid.

i) An internal DTD declaration:

If the DTD is declared inside the XML file, it must be wrapped inside the `<!DOCTYPE>` definition:

XML document with an internal DTD

```
<?xml version="1.0"?>
```

```
<!DOCTYPE note [
```

```
<!ELEMENT note (to, from, heading, body)>
```

```
<!ELEMENT to (#PCDATA)>
```

```
<!ELEMENT from (#PCDATA)>
```

```
<!ELEMENT heading (#PCDATA)>
```

```

<! ELEMENT body ( # PCDATA ) >
] >
<note>
  <to> Tour </to>
  <from> Jani </from>
  <heading> Remiden </heading>
  <body> don't forget in this weekend </body>

```

(ii) An external DTD declaration

If the DTD is declared in an external file, the `<!DOCTYPE>` definition must contain a reference to the DTD file:

XML document with a reference to an external DTD:

```

<?xml version = "1.0"?>
<!DOCTYPE note SYSTEM "note.dtd">
<note>

```

```

  <to> Tour </to>
  <from> Jani </from>
  <heading> Remainder </heading>
  <body> don't forget in this weekend! </body>

```

</note>

and here is the file "note.dtd", which contain the

```

DTD: <!ELEMENT note (to, from, heading, body)>
<!ELEMENT to ( # PCDATA )>
<!ELEMENT from ( # PCDATA )>
<!ELEMENT heading ( # PCDATA )>
<!ELEMENT body ( # PCDATA )>

```

2. Explain in detail about XML schema elements and attributes.

XML schema elements

- XML schemas define the elements of your XML files.
- A simple element is an XML element that contains only text. It cannot contain any other elements or attributes.

Defining a simple element

The syntax for defining a simple element is:

`<xs:element name = "xxx" type = "yyy" />`

When, xxx is the name of the element

Yyy is the data type of the element

XML schema has a lot of built-in data types.

- ✓ xs:string
- ✓ xs:decimal
- ✓ xs:integer
- ✓ xs:boolean
- ✓ xs:date
- ✓ xs:time

eg: `<lastname>Ref shes</lastname>`
`<age>36</age>`
`<date born>1970-03-27</date born>`

Default and Fixed:

- Default value: It is automatically assigned to the element when no other value is specified. In the following example the default value is 'red'.

`<xs:element name = "color" type = "xs:string" default = "red">`

XSD attributes

- Simple elements cannot have attribute. If an element has attributes, it is considered to be of a complex type.
- But the attribute itself is always declared as a simple type.

Syntax

<xs:attribute name="xxx" type="yyy"/>

Eg: An XML element with an attribute:

<lastname lang="EN">Smith</lastname>

The corresponding attribute definition:

<xs:attribute name="lang" type="xs:string"/>

Default and Fixed value for attributes:

Attributes may have a default value OR a fixed value specified.

→ default value: It is automatically assigned to the attribute when no other value is specified.

In the following example the default value is "EN":

<xs:attribute name="lang" type="xs:string" default="EN"/>

• Fixed value: It is also automatically assigned to the attribute, and you cannot specify another value. In the following example value is "EN".

<xs:attribute name="lang" type="xs:string" fixed="EN"/>

Optional and Required attribute

• Attributes are optional by default.

• To specify that the attribute is required, use the 'use' attribute.

<xs:attribute name="lang" type="xs:string" use="required"/>